DO NOW — RECALL

Operating-system types

Name: _____ Class: _____ Date: _____

1. Which scheduler uses a fixed quantum?

2. What does starvation mean?

3. What matters most to a real-time task?

4. Rising-edge flip-flop Q=0. D pulses high between edges but is 0 at next rising edge. What is new Q?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Round Robin.

2. Model answer

A task can be indefinitely denied execution.

3. Model answer

Meeting the specified response deadline.

4. Model answer

Q remains 0; only D at the active clock edge is sampled.

Vocabulary — know these terms

real-time — Deadline-bound response.

embedded — Dedicated device software.

distributed — Coordinated networked machines.

multi-user — Account-based sharing.

multitasking — Several processes progressing.

NEW LEARNING

Read, annotate and explain

Understand

Real-time OS correctness depends on deadlines, not just average response speed.

Trace

Embedded and distributed refer to system architecture and purpose; each can overlap with real-time.

Evaluate

Multi-user concerns people/account isolation, whereas multitasking concerns processes.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A safety-critical industrial controller has a fixed response deadline. Which OS type suits it?

Predict first

Predictable bounded response is required to meet safety deadlines.

Model and check

Explain why each value follows from the earlier step.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Match the five OS categories to an identifying property. [5]

Question 2

Explain OS requirements of a dedicated emergency-stop controller. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why a networked cloud service can be distributed, multi-user and multitasking. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An embedded OS can never also be real-time.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.